MoCG: Modality Characteristics-Guided Semantic Segmentation in Multimodal Remote Sensing Images

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***Abstract*— The rapid development of satellite platforms has yielded copious and diverse multisource data for earth observation, greatly facilitating the growth of multimodal semantic segmentation (MSS) in remote sensing. However, MSS also suffers from numerous challenges: 1) existing inherent defects in each modality due to the different imaging mechanisms; 2) insufficient exploration of the intrinsic characteristics of modalities; and 3) the existence of the huge semantic gap between heterogeneous data causes difficulties in feature fusion. The inability to effectively utilize the rich and diverse information provided by each modality and ignorance of the heterogeneity between modalities will hinder the feature enhancement, and further significantly impacts the semantic segmentation accuracy. Furthermore, neglecting the huge gap makes feature fusion challenging. In this study, we introduce a novel framework for MSS that effectively mitigates the aforementioned problems. Our approach employs a pseudo-Siamese structure for feature extraction. Specifically, we propose a simple yet effective geometric topology structure modeling (GTSM) module to extract geometric relationships and texture information from optical data. Additionally, we present a modality intrinsic noise suppression (MINS) module to fully exploit radiation information and alleviate the effects of unique geometric distortions for synthetic aperture radar (SAR). Furthermore, we present an adaptive multimodal feature fusion (AMFF) module for fully fusing different modality features. Extensive experiments on both WHU-OPT-SAR and DFC23 datasets validate the robustness and effectiveness of the proposed Modality Characteristics-Guided (MoCG) semantic segmentation network compared to other state-of-the-art semantic segmentation methods, including multimodal and single-modal approaches. Our approach achieves the best performance on both datasets, resulting in mean intersection over union (mIoU)/overall accuracy (OA) gains 69.1%/87.5% on WHU-OPT-SAR and 86.7%/97.3% on DFC23.**

***Index Terms*— Cross-modal feature fusion, multimodal semantic segmentation (MSS), optical, remote sensing, synthetic aperture radar (SAR).**

I. Introduction

MULTIMODAL semantic segmentation (MSS) is a crucial task in remote sensing image interpretation, involving the assignment of semantic labels to each pixel in an image utilizing data from different modalities [1]. Compared with single-modal semantic segmentation, MSS leverages complementary information [2], [3] from different modalities showing better performance, and has been widely used in land use classification [4], [5], urban planning [6], [7], agriculture, and ecology [8]. This article specifically focuses on the fusion of optical and synthetic aperture radar (SAR) modalities for semantic segmentation.

The maturity of imaging equipment has brought more possibilities for optical and SAR image interpretation. Due to the significant differences in imaging mechanisms, optical and SAR images show heterogeneous properties. On the one hand, SAR can penetrate clouds and rain, enabling all-weather and all-time observation without being affected by environmental factors. What's more, SAR images exhibit distinct scattering characteristics for different types of ground objects. Ma et al. [9], [10] integrate the superpixel generating and merging processes into an end-to-end trainable process, improving the accuracy of superpixels and thus promoting the performance of SAR image interpretation. Yue et al. [11] propose a new semi-supervised SAR automatic target recognition algorithm in combination with LDA. Nevertheless, SAR data is also accompanied by speckle noise and geometric distortions, including layover and foreshortening [as shown in the first two columns of Fig. 1(a)], which are harmful for Image interpretation. On the other hand, optical images provide an authentic and visually intuitive representation of terrain objects. However, they are susceptible to cloud occlusion and lighting conditions, leading to the loss of contour and detailed information [as shown in the third and fourth columns of Fig. 1(a)]. In a word, optical and SAR images possess distinct strengths and weaknesses. Leveraging the complementary nature of both modalities is crucial for achieving comprehensive and accurate analysis. Therefore, how to

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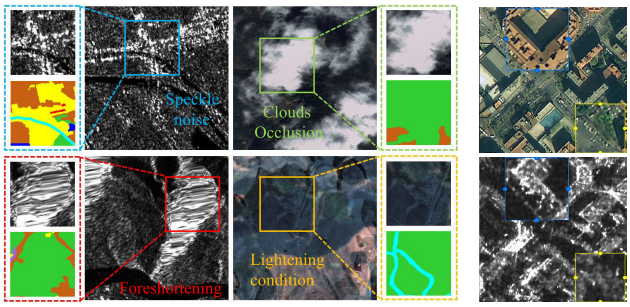


Fig. 1. (a) Illustrates the limitations of optical and SAR images. (1) Speckle noise in SAR images (blue dashed box). (2) Foreshortening in SAR images (red dashed box). (3) Cloud occlusion in optical images (green dashed box). (4) Lightening influence in optical images (yellow dashed box). (b) Shows a couple of optical and SAR images of the same scene. Optical images present ground targets' texture, color, and other features realistically and intuitively. However, SAR generally uses grayscale values to reflect information; the higher the surface reflection intensity, the higher the value.

effectively harness the valuable information of heterogeneous data acquired from various sensors while mitigating inherent noise interference for better segmentation is promising.

With the flourishing development of deep learning, numerous studies [1], [2], [3], [12], [13], [14], [15], [16] have provided evidence that multimodal fusion can enhance semantic segmentation performance. According to the location where the fusion occurs, multimodal fusion techniques can be categorized into three main types: data fusion, early fusion, and late fusion [2]. Data fusion [17] involves directly merging the original input images, which is a straightforward technique and commonly used for homogeneous data. Early fusion, also referred to as feature fusion [18], [19], [20], [21], takes place after extracting features. Late fusion [22], [23], also known as decision-level fusion, directly fuses segmentation results obtained from both modalities. Early fusion and late fusion are commonly employed techniques when dealing with heterogeneous data, such as optical and SAR data. Simple feature fusion methods, such as concatenation along the channel dimension or direct addition of features from different modalities, do not adequately account for the semantic gap, further restricting the utilization of supplementary information.

In recent years, to overcome these limitations, attention-based fusion models have been proposed for effectively integrating unique features obtained from optical and SAR images [24], [25], [26], [27]. These models can fully capture the salient features by focusing on regions of interest. Moreover, CFNet [28], MFNet [29], CMGFNet [30], and CEGFNet [31] leverage gate mechanisms to eliminate redundant features and suppress interference, improving the effectiveness of feature fusion. In addition, transformer [32], [33], which possesses powerful capabilities for capturing long-range dependencies, has also been introduced into the field of MSS for improved extraction [34], [35] or fusion [36], [37] of modality-specific features. Although these methods have improved segmentation accuracy to some extent, there still exist the following challenges in remote sensing:

A. Lack of Exploration of the Modality-Specific Characteristics

Generally speaking, due to the different imaging mechanisms, optical and SAR images have distinct defects [as shown in Fig. 1(a)]. What's more, they have different representations for the same scene [as shown in Fig. 1(b)]. The utilization of a common backbone in current methods for extracting features from each modality hinders the complete extraction of complementary information, thereby limiting the semantic segmentation accuracy.

B. Existence of Huge Semantic Gap Between Modalities

Neglecting the semantic gap and directly conducting feature fusion make it difficult for model convergence, and it cannot utilize complementary advantages to make up for the limitations of individual modalities. Consequently, it is crucial to introduce an innovative fusion strategy that effectively mitigates the influence caused by the semantic gap.

To address the abovementioned challenges, we introduce an innovative framework for MSS that can fully exploit and fuse the modality-specific features effectively. Specifically, this network consists of several key components, including a hierarchical geometric topology structure modeling (GTSM) module in the optical branch, a pyramid modality intrinsic noise suppression (MINS) module for SAR feature extraction as well as an adaptive multimodal feature fusion (AMFF) module at multiscales with similarity constraint.

Specifically, to fully explore the modality-specific characteristics and alleviate the impact of intrinsic attributes, GTSM models the ground objects as a series of graph nodes allowing for the comprehensive utilization of spectral and texture information from optical images, as well as the exploration of invariant topological structures among objects. What's more, the MINS module regards optical features as guidance to refine the SAR features based on an improved transformer, which makes full use of radiation information while mitigating intrinsic defects arising from the imaging mechanism. Furthermore, to maximize the utilization of the complementary information across different modalities, we mitigate the differences between features of heterogeneous data by constraining features at multiscales using knowledge distillation. Fig. 2 illustrates the MSS pipeline between our proposed **Mo**dality **C**haracteristics-**G**uided (MoCG) and other approaches.

Extensive experiments and comprehensive evaluations clearly demonstrate that our MoCG framework can effectively extract and fuse features within modalities, significantly reducing the influence of semantic differences between modalities. Our approach achieves superior performance on both WHU-OPT-SAR and DFC23 [38] datasets and surpasses other SOTA methods in effectiveness and robustness. The key innovations and contributions of this study are summarized as follows.

1) This study analyses the potential issues of optical and SAR images and the challenges of MSS in remote sensing, then proposes a novel network named MoCG semantic segmentation in multimodal remote sensing images, which fully capturing the inherent characteristics and effectively fusing features for better segmentation.

2) In order to extract optical-specific structural and textural properties in optical images, the **GTSM** module models the topological relationships among objects by transforming the image features into graph-structured representation. Besides,

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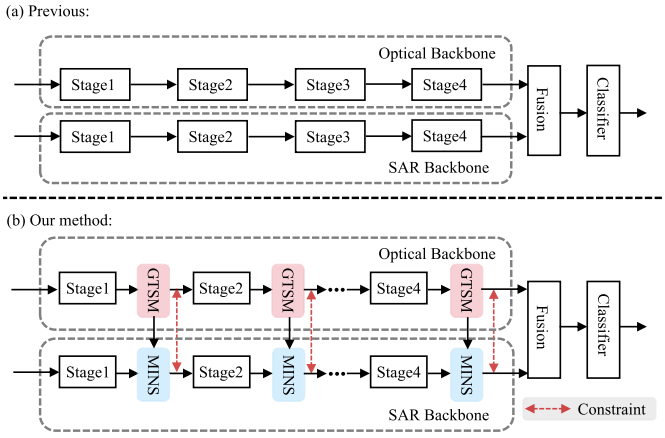


Fig. 2. MSS pipeline between our proposed MoCG and other methods. (a) In previous works, the same backbone is often used to extract features for optical and SAR images. Then they conduct fusion to integrate the complementary information using attention, gate, or transformer. (b) Pipeline of our proposed method. We design modality-specific backbones to obtain inherent characteristics of different modalities with the GTSM module and the MINS module. Moreover, we use KL divergence to constrain the similarity distribution between two modalities in a multiscale.

the **MINS** module is proposed to explore the various scattering information and prevent the interference of intrinsic defects under the guidance of optical images.

3) For alleviating the impact of the semantic gap and fully utilizing the useful information in both modalities, we introduce the **AMFF** module. It adaptively fuses the complementary information from two modalities and avoids the feature mismatch caused by semantic gap through constraining the similarity affinities, thus improving the semantic segmentation performance.

The subsequent sections of this article are structured as follows. Section II provides an overview of relevant literatures. In Section III, we present a comprehensive explanation of our proposed method. Subsequently, Section IV encompasses the demonstration of our experiments, including ablation studies, comparative experiments, visualization results, and corresponding analysis. Finally, Section V concludes by summarizing our key points and offering a glimpse into future prospects.

II. Related Works

We primarily introduce the semantic segmentation in natural scenes in Section II-A. Then, in Section II-B, we briefly introduce semantic segmentation in remote sensing. Subsequently, we provide an overview of various MSS methods in Section II-C.

A. Semantic Segmentation in Natural Scenes

Semantic segmentation, a critical task in image interpretation, has made remarkable advancements, primarily attributed to the utilization of powerful neural networks pre-trained on large datasets [39]. Through the substitution of fully connected layers with convolutional layers, FCN-based methods [40], [41], [42], [43], [44] have achieved notable progress and breakthroughs in semantic segmentation. After that, Unet [45] and SegNet [46] adopt an encoder–decoder architecture that combines shallow features, which preserve high spatial resolution, with deep features that contain rich semantic information. Nevertheless, these methods still face challenges in adequately capturing long-range relationships. To overcome this issue, the Deeplab series [42], [43], [47] utilize dilated convolutions to enlarge receptive fields. Similarly, PSPNet [44] incorporates a pyramid pooling module (PPM) to capture contextual information across multiple scales.

Additionally, attention mechanisms [48], [49], [50], [51] have gained widespread adoption in semantic segmentation, which further improves the segmentation performance by fully modeling contextual information. Furthermore, with the remarkable success of vision transformers (ViT) [52] in image classification, there has been an increasing interests in applying transformer-based models to various computer vision tasks, including semantic segmentation. As a result, recent methods [53], [54], [55], [56], [57], [58] have explored incorporating transformers as encoder, decoder, or both to enhance the accuracy of semantic segmentation through better capturing global contextual information. Moreover, StructToken [56] proposes a new paradigm for semantic segmentation. It utilizes structural information as a priori to directly constructing semantic masks, bypassing the need for per-pixel classification.

B. Semantic Segmentation in Remote Sensing

In contrast to natural scenes, remote sensing data presents a more complex scenario characterized by the presence of numerous small objects as well as foreground–background imbalance challenges. Many scholars have proposed various methods to tackle these issues in recent years. For instance, FarSeg [59] introduces a foreground-aware framework that models foreground from relation-based and optimization-based perspectives to mitigate foreground–background imbalance and false alarm problems. Niu et al. [60] introduce a pyramid graph unit and disentangled learning-based segmentation framework that explicitly models foreground and boundary objects, capturing effective global inter-dependencies and addressing foreground–background imbalance and significant intra-class variance issues in aerial imagery. FactSeg [61] presents a foreground activation-driven network that tackles the challenge of accurately segmenting small objects. The method incorporates self-organizing maps (SOMs)-based optimization, allowing for effective handling of small objects and improving the overall segmentation performance. Feng et al. [62] introduce an edge-weighted affinity loss to improve the segmentation accuracy of small objects and boundaries. Additionally, [63], [64], [65] devote to classifying the boundary pixels, trying to address the issue of boundary blur.

C. Multimodal Semantic Segmentation

As the capabilities of single sensors in capturing comprehensive features of objects in remote sensing imagery are limited, there has been an increasing interest in utilizing data with multimodal characteristics to enhance image interpretation. Many scholars have focused on research into multimodal fusion methods [1], [2], [3], [12], [13]. The

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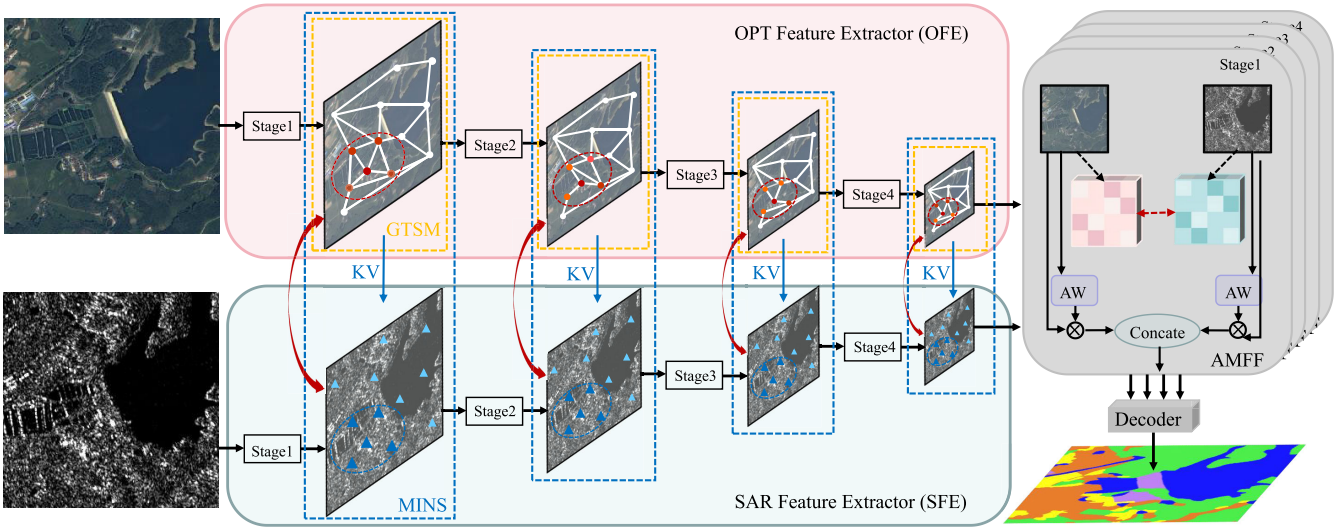


Fig. 3. Overall pipeline of our proposed MoCG. The GTSM module is integrated in each stage of the OFE. The MINS module is plugged into each stage of the SFE. "⊕" denotes the element-wise summation and "‖" represents the concatenation operation. "AW" refers to the adaptive weighting module.

simplest fusion methods involve direct summation or concatenation. For example, methods such as PSCNN [18] and MRSDC [19] concatenate features along the channel dimension, while FuseNet [20] and vFuseNet [21] adopt addition operations to fuse features from different modalities. To further enhance feature fusion, attention-based networks have been proposed. MBFNet [26] introduces a bilinear attention module using global average pooling and global max-pooling to perform fusion. AFNet [24] and CroFuseNet [27] also incorporate channel attention mechanisms to improve the fusion of optical and SAR features for road extraction. Similarly, MCANet [25] proposes a multimodal-cross attention module to further boost semantic segmentation performance.

Some studies have introduced gate mechanisms to suppress redundant information and improve the efficiency of multimodal fusion. LIGHT [66] designs a gated interaction module to perform feature interactions between branches efficiently. Additionally, CMGFNet [30] and CEGFNet [31] both incorporate the gate mechanism into their model for effective fusion. Recently, the transformer architecture has been introduced into the field of multimodal fusion as well. CMFNet [37] proposes a multiscale cross-modal transformer-based feature fusion method. Besides, CEN [58] presents a parameter-free channel switching network that adaptively exchanges channels between different modality sub-networks. Additionally, TokenFusion [35] dynamically detects uninformative tokens and replaces them with projected and aggregated cross-modal features for efficient feature fusion. In the medical imaging field, mmFormer [34] proposes a new multimodal medical transformer for incomplete multimodal learning to address cases where complete MRI scans are unavailable.

Although multimodal data provides substantial complementary information to enhance image interpretation in remote sensing, it also poses significant challenges [12], [13]. The diverse imaging mechanisms of various remote sensing devices capture complex object characteristics with a large semantic gap between modalities, making it difficult to map multimodal data to the same feature space for stable and effective feature fusion. Therefore, it is essential to effectively extract modality-specific characteristics, reduce the influence of semantic gap between different modalities, and efficiently integrate complementary information. In the study, we propose a method that combines graph convolution and cross-transformer to extract optical and SAR's intrinsic characteristics. Subsequently, we adopt a novel fusion module that narrows the gap between modalities and ensures a high-quality fusion of complementary information.

III. Method

We first introduce the overall pipeline of our proposed MoCG in Section III-A. Subsequently, we provide a detailed introduction of the GTSM module, the MINS module, and the AMFF module. Finally, we offer a discussion on the selection of loss functions.

A. Overall Framework

Fig. 3 illustrates the overall framework of our proposed MoCG, which is successively composed of pseudo-Siamese feature extractors (ResNet-50 [67]/RegNet-3.2 [68]), AMFF modules and the FPN-based decoder. Two branches of the pseudo-Siamese backbone both consist of four stages. The optical branches are equipped with the GTSM modules and the SAR branches comprise the MINS modules between every two stages.

Specifically, the input optical images and SAR images are fed into the optical feature extractor (OFE) and SAR feature extractor (SFE), respectively. The GTSM modules are inserted into OFE to explore unique topological and structural relationships between objects and dig out multiscale more discriminative features. In addition, the MINS modules are plugged into SFR to dig out the specific scattering characteristics of ground objects and mitigate the impact of SAR intrinsic defects under the guidance of optical images. Next,

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the features from both branches are sent into the AMFF modules based on similarity constraint aim to effectively integrate complementary information provided by two modalities at multiscale. After that, a top-down path taking the four fully fused features as input is adopted for recovering the detailed information. Finally, the output feature is refined through a ConvBlock to obtain the final segmentation map.

Algorithm 1 Workflow of the Proposed MoCG

Input:
Input optical image: $X \in R^{3\times H\times W}$
Input sar image: $Y \in R^{3\times H\times W}$
Label: $L \in R^{H\times W}$
T_{train}: training phase
Output:
THe semantic segmentation results: $O_{final} \in R^{H\times W}$

while T_{train} **do**
  Extract Features:
  OPT branch: Using **GTSM** to model the topology relationships, obtain hierarchical features $F_{opt_i}, i = 1, \ldots, 4$
  SAR branch: the **MINS** module takes OPT features F_{opt_i} and SAR features F_{sar_i} as input to get refined SAR features $F_{sar_i}, i = 1, \ldots, 4$
  Feature fusion:
  AMFF module takes F_{opt_i} and F_{sar_i} of the same stage as input to obtain fused features.
  First, use the KL divergence to constraint the similarity affinities:$S_{opt}, S_{sar} \in R^{HW\times HW}$
  Then, generate data-driven weights $\omega_{opt}, \omega_{sar}$ to adaptive Obtain the fused features
  $F_i = \sigma(\omega_{sar_i} \cdot F_{sar_i} \parallel \omega_{opt_i} \cdot F_{opt_i})$
  Optimization:
  Calculate the Loss function of MoCG:
  Calculate the segmentation loss: L_{ce} (Eq. 13)
  Calculate the auxiliary loss: L_{aux} (Eq. 13, 19, 20)
  Calculate the KL Divergence loss: L_{KL} (Eq. 16)
return O_{final}

B. Hierarchical GTSM Module

To acquire optical-specific geometric and structural features of objects and obtain the relative position relationship among ground objects that do not change with time and other factors, we leverage graph convolution to model the unchanged topology relationships among objects. The diagram of the complete GTSM module is presented in Fig. 4.

We first transform the features from Euclidean space into graph representations in non-Euclidean space by modeling the objects as vertices. The goal of this is to group adjacent pixels with similar features into the same vertex, corresponding to the same object region. Specifically, for the single-modal optical image features, we convert the feature $X \in R^{C\times H\times W}$ into a graph representation $G \in R^{N\times D}$, D denotes the feature dimension of the node and N is a hyperparameter, representing the number of nodes.

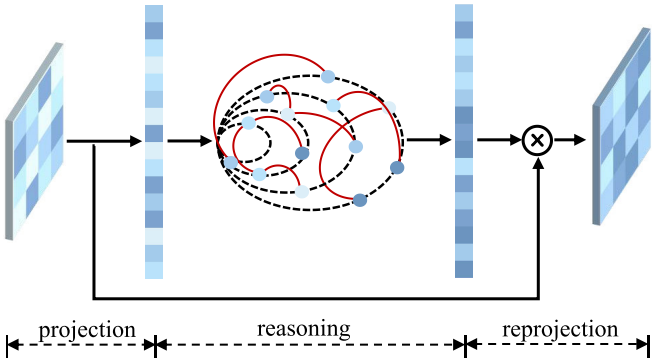


Fig. 4. Diagram of our proposed GTSM module. First, we project the image features into graph nodes based on their similarities. Next, we apply two sequential graph convolution layers to model topological relationships between objects. Finally, we reproject the reasoned node features into coordinate space as a global vector and multiply it with the original input.

It is worth noting that shallow features tend to focus on detailed information, such as texture and color, and have small receptive fields. In contrast, deep features emphasize semantic information, such as categories with large receptive fields. Therefore, we construct a hierarchical graph structure to capture finer object relationships. For example, we model each object as an individual graph node for shallow features to establish object–object dependencies. As the network becomes deeper, we gradually merge object entity nodes, and the node will include more semantic information. Therefore, we can gradually capture rich details and semantic information from shallow to deep.

To propagate information in non-Euclidean space, we employ two consecutive graph convolution layers, as expressed in 1. Given the node feature $G \in R^{N\times D}$, $A \in R^{N\times N}$ denotes the trainable adjacency matrix, and $W_1 \in R^{D\times F}, W_2 \in R^{F\times D}$ denote the learnable transformation matrix to control the dimension of the output node features. The simplest graph convolution operation is formulated as follows:

$$G' = \sigma(A_2\sigma(A_1 G W_1)W_2) \tag{1}$$

where $\sigma(\cdot)$ represents the ReLU activation function and G' denotes the feature obtained from the graph convolution.

At last, we should reproject the graph node features into coordinate space. Specifically, we directly review the node feature into a global vector and multiply it with the input features with channel dimension was changed by one convolution layer to adaptive with the output features' channel dimension.

1) Hierarchical GTSM in Optical Feature Extraction Branch: As illustrated in Fig. 4, the input features' channel of F_1 to F_4 is 96, 192, 432, and 1008, respectively. First, we use 1×1 conv to normalize the feature dimension of each stage into 256 and then plug them into the GTSM module. Combining with the previous analysis in this section, we set the number of nodes to 64, 32, 16, and 8 for each stage in the experiment. Intuitively, the hierarchical geometric structure topology modeling module can effectively capture not only detailed information but also semantic relation between objects while taking multiscale information into account.

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C. MINS Module

We integrate a transformer-based structure to tackle the problem of speckle noise-induced erroneous information and overcome the challenges posed by inherent properties such as foreshortening in SAR images. This structure guides and corrects SAR features using the complementary information provided by optical features. What's more, by leveraging the spectral information present in SAR data, we aim to compensate for the limitations of optical images, which are susceptible to weather conditions.

The standard transformer takes a 1-D sequence as input. In order to adapt to this form, the vision Transformer first splits an image into a sequence of flattened 2-D patches. A transformer encoder, comprising multihead attention, MLP blocks, and LayerNorm, is then employed to operate on these embeddings. To leverage the supplementary information from different modalities, we extend and introduce a cross-information-guided transformer for SAR feature refinement.

We first split 2-D optical images $X \in R^{C\times H\times W}$ and 2-D SAR images $Y \in R^{C\times H\times W}$ into non-overlapping patches, like ViT [52]. Each patch is treated as a token, let the patch size is $p \times p$, then the token embeddings of both inputs are $X_p \in R^{N\times(p^2\times C)}$, $Y_p \in R^{N\times(p^2\times C)}$, where $N = (HW/p^2)$ represents the number of tokens, $p^2 \times C$ stands for the feature dimension of each token. Similar to ViT, we use a trainable 1-D position embeddings to retain the location information. Finally, we apply a linear layer to map the feature dimension into an arbitrary dimension (denoted as D). After projecting, we get two token embeddings $X_i \in R^{N\times D}$, $Y_i \in R^{N\times D}$

$$X_i = \text{Linear}\left(X_p\right) + E_{\text{pos}} \tag{2}$$

$$Y_i = \text{Linear}\left(Y_p\right) + E_{\text{pos}} \tag{3}$$

where Linear $\in R^{p^2C\times D}$ represent the linear map layer, and $E_{\text{pos}} \in R^{N\times D}$ represent the position embeddings.

After tokenizing the images, the cross-transformer encoder will simultaneously be applied to the optical and SAR patch embedding. Specifically, we replace the conventional MSA module within a transformer block with a multihead cross-attention (MCA) module to construct the cross-transformer. As illustrated in Fig. 5, a cross-transformer block comprises a MCA module, which is subsequently followed by an MLP block. Before each block, we employ the LayerNorm (LN), and after each block, residual connections are utilized.

Cross-attention takes the SAR embeddings and optical embeddings as inputs, where SAR embeddings serve as query (Q) and optical embeddings act as key (K) and value (V)

$$Q = \text{Linear}(Y_i) \in R^{N\times D} \tag{4}$$

$$K = \text{Linear}(X_i) \in R^{N\times D} \tag{5}$$

$$V = \text{Linear}(X_i) \in R^{N\times D}. \tag{6}$$

Utilizing SAR features to query useful information in optical features to mitigate the influence of SAR inherent defects and correct the wrong information caused by speckle noise, foreshortening and so on

$$\text{MCA} = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \times V. \tag{7}$$

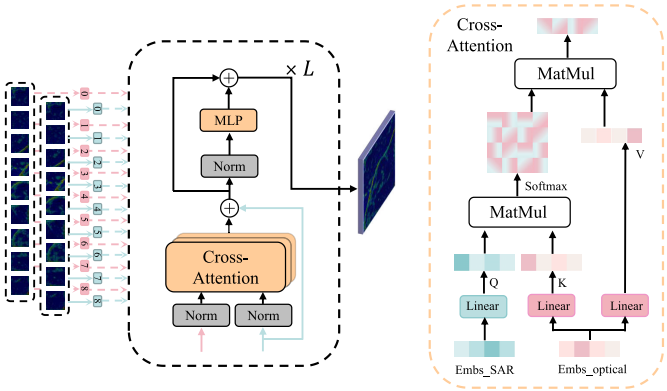


Fig. 5. Diagram of our proposed MINS module. The MINS module takes optical and SAR features as input simultaneously and transforms them into flattened tokens, respectively. Then, SAR tokens are treated as query and optical tokens as key and value, using optical features to refine SAR images and correct the wrong information caused by speckle noise and geometric distortion in SAR images.

After MCA, the SAR features have been modified by optical features

$$Y_i' = \gamma * \text{MCA} + Y_i. \tag{8}$$

Finally, the MLP will be applied on the modified features, which is formulated as follows:

$$Y_{\text{out}} = Y_i' + \text{MLP}\left(Y_i'\right). \tag{9}$$

Noting that $LN()$ is ignored in the formula.

Due to the significant variation of object size in remote sensing, we design the hierarchical structure to capture the multiscale information. With the network deeper, we retain the patch size and reduce the number of patches. In this study, the number of patches is 64, 64, 16, and 16, respectively.

1) Hierarchical MINS in SAR Feature Extraction Branch: Similar to hierarchical GTSM, we plug this module between stages in the SAR feature extraction branch to refine the SAR feature gradually.

D. AMFF Module

Capturing modality-specific intrinsic characteristics is a fundamental requirement for MSS tasks. Reducing the impact of semantic gaps between heterogeneous data and fully integrating the complementary information from different modalities are also crucial for achieving high segmentation performance. Despite the vast semantic differences between SAR and optical data, we observe that the similarity relationships of the two modalities should exhibit no significant divergence. Specifically, for both modalities, the affinity within the same class exhibits higher similarity, while that across different classes displays lower similarity. Leveraging this property, we introduce an AMFF module incorporating constraints based on similarity distribution. In detail, for optical feature map $X \in R^{C\times H\times W}$ and SAR feature map $Y \in R^{C\times H\times W}$, we calculate the normalized cosine similarity at the first step, obtaining $S_x \in R^{HW\times HW}$ and $S_y \in R^{HW\times HW}$

$$S = \frac{f(x_i)\cdot f\left(x_j\right)}{\| f(x_i) \|_2^2 \cdot \| f\left(x_j\right) \|_2^2}. \tag{10}$$

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From a statistical perspective, the Kullback–Leibler divergence quantifies the dissimilarity between two probability distributions. When two distributions are identical, the KL divergence is equal to 0. Conversely, the greater the difference between the two distributions, the greater the value of KL divergence.

Assuming the true distribution is $P(x)$ and the predicted distribution is $Q(x)$, we can calculate the KL divergence as follows:

$$\mathrm{KL}(P \parallel Q) = \sum_{x \in X} P(x) \log \frac{P(x)}{Q(x)}. \tag{11}$$

Taking this into consideration, we choose KL divergence to measure the difference in similarity distributions, trying to narrow the divergence in optical and SAR modalities. Except for narrowing down the gap, suppressing the redundant information in different also makes sense in MSS. Hence, we learn the data-driven weights ω_x for optical and ω_y for SAR to fuse their features adaptively

$$F_i = \sigma\left(\omega_x \cdot F_{xi} \parallel \omega_y \cdot F_{yi}\right) \tag{12}$$

where F_i represents the ith fusion feature after stage i.

E. Loss Function

According to our design, we employ three kinds of loss function: segmentation, KL, and auxiliary losses.

1) Segmentation Loss: Given that L_{ce} represents the multiclass cross-entropy loss function

$$L_{ce} = -\frac{1}{N} \sum_{i} \sum_{c=0}^{C-1} y_{ic} \cdot \log(p_{ic}) \tag{13}$$

$$y_{ic} = \begin{cases} 1, & \text{if } i \in c \\ 0, & \text{others} \end{cases} \tag{14}$$

where C represents the number of categories, y_{ic} is standing for the sign function, it equals 1 when the ith pixel belongs to the class, and it equals 0 in other cases. In the experiments, we use standard CE loss as the final seg supervision

$$L_{\mathrm{seg}} = L_{ce}. \tag{15}$$

2) KL Divergence: We utilize the KL divergence to measure the difference of similarity affinities in our experiments

$$L_{\mathrm{KL}} = \left\{\phi(\cos(y_c^{\mathrm{sar}}), \phi(\cos(y_c^{\mathrm{opt}})\right\} = \mathrm{KL}\left(\phi\left(S^{\mathrm{sar}}\right), \phi\left(S^{\mathrm{opt}}\right)\right) \tag{16}$$

where $\cos(\cdot)$ is defined as pixel affinity map in (10) and $\phi(\cdot)$ is defined as transforming the probability to the distribution in (17)

$$\phi(y_c) = \frac{\exp\left(\frac{y_{c,i}}{\tau}\right)}{\sum_{i=1}^{HW} \exp\left(\frac{y_{c,i}}{\tau}\right)} \tag{17}$$

$$\mathrm{KL}\left(y_{\mathrm{sar}}, y_{\mathrm{opt}}\right) = \frac{\tau^2}{C} \sum_{c=1}^{C} \sum_{i=1}^{HW} \phi\left(S_{c,i}^{\mathrm{opt}}\right) \cdot \log \frac{\phi\left(S_{c,i}^{\mathrm{opt}}\right)}{\phi\left(S_{c,i}^{\mathrm{sar}}\right)}. \tag{18}$$

Noting that KL divergence is an asymmetric measure. In the experiment, we use optical features to guide the SAR branch.

3) Auxiliary Loss: To enhance performance and accelerate model convergence, we incorporate additional supervision in the third stage of the RegNet for both branches, like [44]. In the optical branch, we employ a standard cross-entropy loss. However, due to foreshortening in SAR images, adjacent pixel classes often represent pixels that should be assigned to one class, leading to reduced prediction accuracy. Thus, we adopt the multiclass focal loss [69] as the supervisory signal for the SAR branch

$$L_{\mathrm{FL}} = -\frac{1}{N} \sum_{i} \sum_{c=0}^{C-1} y_{ic} \cdot (1 - p_{ic})^{\gamma} \cdot \log(p_{ic}). \tag{19}$$

In the experiment, γ is assigned as 2. Consequently, the whole auxiliary loss is defined as

$$L_{\mathrm{aux}} = L_{ce} + L_{\mathrm{FL}}. \tag{20}$$

By combining the three mentioned loss functions, the formulation of the overall loss is as follows:

$$L = L_{\mathrm{seg}} + \alpha L_{\mathrm{aux}} + \beta L_{\mathrm{KL}} \tag{21}$$

where the choice $\alpha = 9$ and $\beta = 0.1$ is the optimal value obtained by the experiments (Section IV-D5).

IV. Experiment

A. Datasets

1) WHU-OPT-SAR: The WHU-OPT-SAR dataset introduced by MCANet [25] is a pioneering multimodal dataset explicitly designed for land use classification and is the largest dataset of its kind. This dataset encompasses 100 pairs of optical and SAR images, and each pair represents the same geographic area. Besides, the images in this dataset have the same spatial resolution of 5 m and dimensions of 5556 × 3704 pixels. Moreover, this dataset comprises seven primary land use categories, namely farmland, city, village, water, forest, road, and others. To align with the official dataset guidelines, we crop the dataset into 256 × 256 pixel patches without overlapping. Subsequently, the dataset is split into three subsets: a training set containing 17 640 images, a validation set containing 5880 images, and a testing set comprising 5880 images.

2) DFC23: The DFC23 [38] is a large-scale, fine-grained, multimodal benchmark for classifying building roof types that include both optical and SAR imagery. This dataset is organized into 12 categories based on roof geometry, namely flat_roof, gable_roof, gambrel_roof, row_roof, multiple_roof, hipped_roof_v1, arched_roof, hipped_roof_v2, mansard_roof, pyramid_roof, arched_roof, dome, and others. Notably, the DFC23 dataset was specifically curated for the 2023 IEEE GRSS Data Fusion Contest. Following the official dataset settings, we convert the 12-category instance segmentation into 12-category semantic segmentation. The dataset is randomly partitioned during the experiments into three parts for training, validation, and test. According to the official guidelines, the partitioning ratio is 7:2:1.

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TABLE I
COMPARISONS WITH MULTIMODAL STATE-OF-THE-ART ON WHU-OPT-SAR DATASET

Method	Modality	Backbone	UA per category(%)							OA(%)	Kappa(%)	mIoU(%)	mF1(%)
			farmland	city	village	water	forest	road	others				
CNN-based Methods													
PSCNN [18]	OPT&SAR	—	81.5	63.9	56.2	73.5	90.1	38.0	20.1	79.7	71.2	50.7	64.8
CFNet [28]	OPT&SAR	MobileNet-V2 [70]	81.5	80.9	50.5	68.8	90.5	19.9	37.5	79.9	71.3	52.7	66.2
MBFNet [26]	OPT&SAR	ResNet-50 [67]	82.6	78.7	59.9	76.9	90.4	44.0	13.0	81.5	73.9	51.1	64.9
V_FuseNet [21]	OPT&SAR	ResNet-50 [67]	83.9	73.1	60.7	78.5	89.2	53.0	27.2	81.6	74.0	48.2	61.0
MRSDC [19]	OPT&SAR	—	83.5	81.2	57.6	74.1	90.9	29.4	37.4	81.9	74.2	54.2	68.2
DDHRNet [71]	OPT&SAR	HRNet-48 [72]	84.2	69.6	59.7	80.1	89.9	48.1	23.8	82.0	74.4	52.3	66.6
FuseNet [20]	OPT&SAR	ResNet-50 [67]	83.7	68.8	63.1	80.1	89.9	48.1	23.8	82.2	74.5	54.2	63.7
MCANet [25]	OPT&SAR	ResNet-101 [67]	85.1	74.5	65.5	81.6	89.0	54.3	26.7	82.7	75.5	58.3	71.7
Transformer-based Methods													
CMFNet [37]	OPT&SAR	VGG16 [73]	80.0	66.7	64.3	79.7	90.8	54.3	24.7	81.1	73.4	52.1	66.5
mmFormer [34]	OPT&SAR	—	84.8	81.6	72.8	83.5	87.8	67.9	53.2	83.8	77.2	55.1	67.0
TransFuser [36]	OPT&SAR	ResNet-50 [67]	85.0	84.6	65.2	75.6	91.6	43.2	48.6	83.9	77.3	61.8	74.9
TokenFusion [35]	OPT&SAR	MiT-b1 [53]	84.5	85.5	60.4	80.5	91.2	48.8	37.4	84.0	77.3	62.3	75.0
CEN [58]	OPT&SAR	ResNet-50 [67]	85.4	82.9	64.1	79.3	91.8	43.6	38.5	84.2	77.7	59.8	73.4
Ours	OPT&SAR	ResNet-50 [67]	86.7	**85.1**	71.3	83.8	93.1	67.2	57.0	87.0	81.6	68.0	80.3
Ours	**OPT&SAR**	**RegNet-3.2** [68]	**86.8**	84.8	**73.2**	**85.4**	**93.3**	**73.0**	**60.7**	**87.5**	**82.5**	**69.1**	**81.1**

B. Evaluation Metrics

To thoroughly assess the effectiveness of our proposed MoCG, we perform experiments on two distinct datasets. In order to provide a rigorous and thorough analysis of our presented method, we employ a range of widely recognized evaluation metrics. Consistent with the methodology employed in prior research [25], for the WHU-OPT-SAR dataset, we adopt overall accuracy (OA) and Kappa coefficient (Kappa) as the primary metrics and utilize user's accuracy (UA) for the fine-grained analysis. Notably, UA signifies the proportion of correctly predicted positive instances relative to the total number of positive predictions made. In addition, intersection over union (IoU) measures the degree of overlap between the prediction and ground truth, accounting for both correct and misclassified pixels. The F1 score represents the harmonic mean of accuracy and recall, allowing for a balanced assessment of both accuracy and recall. These two metrics offer a more comprehensive evaluation of semantic segmentation performance. Hence, we also use mIoU and mF1 as the additional metrics during the experiments. As for the DFC23 dataset, we use the more recognized metrics in semantic segmentation task OA and mIoU as the primary metrics and the IoU of each category for the fine-grained analysis.

The definitions of these metrics are as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FN}+\mathrm{FP}} \tag{22}$$

$$F_1 = 2 \times \frac{P \times R}{P + R} \tag{23}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{24}$$

$$\mathrm{Kappa} = \frac{po - pe}{1 - pe} \tag{25}$$

where P represents the precision, R represents the recall, po represents the OA, and the definitions are shown as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{26}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{27}$$

$$po = \mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FN}+\mathrm{FP}} \tag{28}$$

$$pe = \frac{(\mathrm{TN}+\mathrm{FN}) \times (\mathrm{TN}+\mathrm{FP}) + (\mathrm{FP}+\mathrm{TP}) \times (\mathrm{FN}+\mathrm{TP})}{(\mathrm{TN}+\mathrm{TP}+\mathrm{FN}+\mathrm{FP})^2}. \tag{29}$$

C. Implementation Details

1) Baseline Model: We choose the ResNet-50 [67]/RegNet-3.2G [68] as the backbone, which have been pre-trained on ImageNet [39]. In both optical and SAR branches, we replace some standard convolution operations with atrous convolution, and then apply channel concatenation after each stage to perform multiscale multimodal feature fusion. After that, we use a decoder based on FPN to generate the final segmentation results. Compared with our proposed MoCG framework, the baseline model only removes the GTSM modules from optical branches, removes the MINS modules from SAR branches and replaces the feature fusion AMFF modules with the simple channel concatenation operation.

2) Training Setting: To train the network, we utilize SGD optimizer with a momentum of 0.9 and a weight decay of 5e-4. Additionally, we employ the poly learning rate policy, which

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follows the formula $lr = \text{base_}lr \times (1-(\text{iter}/\text{max_iter}))^{\text{power}}$, and the power is set to 0.9. We initialize the learning rate to 0.002. Following the original settings in previous works, we crop the images into 256 × 256 for the WHU-OPT-SAR [25] dataset and 512 × 512 for the DFC23 [38] dataset. Note that due to the different image sizes, we train the network with a batch size of 8 for the WHU-OPT-SAR dataset and 4 for the DFC23 dataset. All the methods are carried out under the PyTorch deep learning framework and conducted on 1× NVIDIA 2080Ti GPU.

In the evaluation stage, we adopt the same patch size as the training setting, and the batch size is set to 1 for both datasets.

To ensure fairness and comparability, all methods are executed under the same experimental conditions.

D. Experiments on WHU-OPT-SAR Dataset

1) Comparative Analysis With Multimodal State-of-the-Art: In this section, we perform comprehensive experiments on the WHU-OPT-SAR validation set to compare our method with existing state-of-the-art multimodal fusion methods. For a more rounded analysis, we compare our MoCG not only with CNN-based approaches but also with transformer-based frameworks. Table I displays the experimental results.

We conduct a comparative analysis between our method with several CNN-based approaches in the first part of Table I. PSCNN [18] adopts late fusion to simply process features from different modalities without using any modality-specific inherent characteristics and achieves the worst results of 79.7% OA. FuseNet [20] employs the simplest fusion approaches by directly adding features, achieving OA of 82.2%, Kappa of 74.5%, mIoU of 54.2%, and mF1 of 63.7%. V_FuseNet [21] attempts late fusion based on FuseNet [20] further obtains OA of 81.6%. Although CFNet [28] combines gate mechanism to filter redundant information, ignoring the impact of the significant semantic gap between two modalities causes mismatched features and leads to poor performance 79.9% in OA. This confirms the second issue listed in Section I, emphasizing the importance of narrowing down the influence of the gap between different modalities. MRSDC [19] follows the concatenation operation after the decoder and takes the essence of different modalities into account. By utilizing different backbones, it achieves better results, including OA of 81.9%, Kappa of 74.2%, mIoU of 54.2%, and mF1 of 68.2%. What's more, DDHRNet [71] with a stronger backbone achieves OA of 82.0%. MCANet [25] introduces a multimodal cross-attention model to fuse different features and gains OA of 82.7%.

Additionally, we compare our framework with some transformer-based approaches, as presented in the second part of Table I. Due to the superior capability of transformer-based approaches in modeling long-range relationships, they outperform CNN-based methods, yielding better results. CMFNet [37] proposes two transformer-based modules named CMtrans and MCATrans to fuse multimodal features and capture global information, which achieves OA of 81.1%. mmFormer [34] is applied in the medical field for brain tumor Segmentation. This method demonstrates robust performance even when one or more modalities are missing, resulting in OA of 83.8%. In TransFuser [36], the authors utilize the transformer to fuse multiscale features from two modalities, gaining OA of 83.9%. Unlike the aforementioned methods, the main idea of TokenFusion [35] is to prune a few single-modal transformer models and use them in multimodal fusion, which reaches OA of 84.0%. Furthermore, the channel-exchanging-network (CEN) [58] demonstrates higher performance in multimodal and multitask dense image prediction. Due to its self-adaptive and parameter-free, the CEN achieves an OA of 84.2%.

Compared with these SOTA researches, our approach demonstrates a satisfactory improvement and achieves the best performance on the WHU-OPT-SAR validation set: OA of 87.6%, Kappa of 82.5%, mIoU of 69.1%, and mF1 of 81.1%.

2) Comparative Analysis With Single-Modal State-of-the-Art: We also conduct a comparative analysis between our MoCG and existing single-modal methods to validate the superiority of multimodal fusion. The overall performance comparison is presented in Table II, with the first part showing the results of CNN-based methods and the second part displaying the results of transformer-based methods. According to Table II, using only optical images as input yields better segmentation performance than using only SAR images. In particular, when only optical images are used as input, the best results are OA of 81.4%, Kappa of 73.7%, mIoU of 54.7%, and mF1 of 68.8%. However, when only SAR images are used as input, the best results are OA of 77.1%, Kappa of 67.4%, mIoU of 43.5%, and mF1 of 57.7%. This highlights the significance of developing distinct feature extractors for individual modalities, as elaborated in Section I. According to Table II, our proposed multimodal framework, which takes both optical and SAR images as input, achieves significant improvements compared to any single-modal method. Thus, it is evident that multimodal approaches can capture more effective information and overcome the limitations of single-modality to achieve better performance.

3) Ablation Analysis for the Proposed Modules: In order to assess the individual contributions of our components, we conduct comprehensive ablation experiments on the WHU-OPT-SAR dataset, and Table III shows the ablation results. Additionally, Fig. 6 presents the UA scores across eight fine-grained categories as each module is incrementally incorporated, and Fig. 7 displays some feature visualizations.

The baseline model achieves suboptimal performance due to the absence of discriminative features across modalities and a lack of effective fusion strategies. Specifically, it obtains OA, Kappa, mIoU, and mF1 of 81.5%, 73.7%, 54.2%, and 68.1%, respectively.

Extracting more discriminative features from different modalities represents a crucial factor in improving semantic segmentation results. By exploring invariant topological relationships, the GTSM module in the optical branch captures structural and geometrical characteristics from optical images, generating finer features with higher inter-class consistency (as shown in the fourth column of Fig. 7) and making significant improvements in all metrics compared with baseline model. Specifically, it achieves a significant 4.5% increase in OA, a notable 6.6% enhancement in Kappa, a remarkable 12.9% improvement in mIoU, and a noteworthy 11.2% boost in

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TABLE II
COMPARISONS WITH SINGLE-MODAL STATE-OF-THE-ART ON WHU-OPT-SAR DATASET

Method	Modality	Backbone	UA per category(%)							OA(%)	Kappa(%)	mIoU(%)	mF1(%)
			farmland	city	village	water	forest	road	others				
CNN-based Methods													
DANet [48]	OPT	ResNet-50 [67]	79.2	71.9	49.7	70.1	89.4	26.9	17.1	78.0	68.7	48.0	61.9
PSPNet [44]	OPT	ResNet-50 [67]	80.8	71.8	57.2	76.2	89.7	43.5	20.9	80.3	72.1	54.4	68.0
Deeplabv3 [43]	OPT	ResNet-50 [67]	80.5	70.5	58.7	76.8	89.5	44.9	23.2	80.3	72.1	54.2	68.0
OCRNet [74]	OPT	HRNet-W48 [72]	82.9	46.5	64.6	77.7	90.5	42.7	41.1	81.0	73.2	49.0	63.7
CCNet [50]	OPT	ResNet-50 [67]	81.4	72.1	61.1	78.3	89.1	48.9	26.8	81.0	73.2	55.9	69.5
UperNet [75]	OPT	ResNet-50 [67]	81.8	70.6	63.1	78.1	90.2	47.8	23.0	81.3	73.6	52.4	66.8
OCRNet [74]	SAR	HRNet-W48 [72]	78.0	65.5	45.0	71.3	72.9	24.8	10.1	70.7	59.4	35.4	47.5
DANet [48]	SAR	ResNet-50 [67]	78.0	67.7	37.3	66.2	87.9	14.5	9.8	75.1	64.3	37.3	49.5
CCNet [50]	SAR	ResNet-50 [67]	78.0	67.2	44.1	71.5	87.9	33.5	16.4	76.6	66.8	42.8	56.9
UperNet [75]	SAR	ResNet-50 [67]	79.9	66.0	43.1	70.7	88.2	27.3	12.9	76.9	67.0	40.1	52.9
PSPNet [44]	SAR	ResNet-50 [67]	78.9	68.2	45.0	70.7	88.1	28.1	15.5	76.9	67.0	41.5	55.0
Deeplabv3 [43]	SAR	ResNet-50 [67]	78.9	68.4	45.4	71.5	87.9	30.7	17.8	77.1	67.4	43.5	57.7
Transformer-based Methods													
SegFormer [53]	OPT	MiT-b1 [53]	79.1	72.9	38.0	64.7	88.1	0.3	0.4	75.5	64.8	40.3	51.7
SETR [56]	OPT	ViT-L [52]	79.0	70.4	47.5	67.8	88.8	17.3	12.1	77.0	67.2	45.2	58.8
Segmenter [54]	OPT	ViT-L [52]	82.3	75.2	51.7	74.4	89.4	16.0	12.0	79.9	71.4	51.2	63.4
TransUnet [57]	OPT	ResNet-50+ViT [52], [67]	81.8	72.3	61.5	78.3	90.1	49.2	23.8	81.4	73.7	54.7	68.8
UnetFormer [55]	OPT	ResNet-50 [67]	80.9	69.4	59.7	76.5	89.8	47.5	21.2	80.3	72.1	49.2	63.7
SegFormer [53]	SAR	MiT-b1 [53]	79.4	65.8	16.2	62.4	86.7	0.0006	0.0005	73.0	60.8	34.2	45.1
Segmenter [54]	SAR	ViT-L [52]	77.3	71.8	22.4	68.2	88.7	0.9	0.2	74.6	63.3	37.9	49.1
SETR [56]	SAR	ViT-L [52]	78.7	65.6	44.2	70.6	87.9	29.6	13.5	76.5	66.4	39.1	51.3
UnetFormer [55]	SAR	ResNet-50 [67]	75.9	79.9	45.0	71.9	88.7	9.8	13.4	76.5	66.8	41.7	54.8
TransUnet [57]	SAR	ResNet-50+ViT [52], [67]	79.9	66.7	43.6	69.7	87.5	31.5	16.1	76.7	66.6	39.8	52.1
Ours	OPT&SAR	ResNet-50 [67]	86.7	**85.1**	71.3	83.8	93.1	67.2	57.0	87.0	81.6	68.0	80.3
Ours	**OPT&SAR**	**RegNet-3.2** [68]	**86.8**	84.8	**73.2**	**85.4**	**93.3**	**73.0**	**60.7**	**87.5**	**82.5**	**69.1**	**81.1**

TABLE III
ABLATION ANALYSIS FOR THE PROPOSED COMPONENTS ON WHU-OPT-SAR DATASET

Method	Module			OA (%)	Kappa (%)	mIoU (%)	mF1 (%)
	GTSM	MINS	AMFF				
Baseline				81.5	73.7	54.2	68.1
Ours	✓			86.0	80.3	67.1	79.3
		✓		86.0	80.3	67.0	79.2
			✓	85.0	78.8	64.3	76.8
	✓	✓		87.0	81.7	68.8	80.7
	✓		✓	86.5	81.0	67.3	79.5
		✓	✓	86.8	81.4	68.0	80.1
	✓	✓	✓	**87.5**	**82.5**	**69.1**	**81.1**

mF1. Furthermore, the fine-grained results presented in Fig. 6 provide further evidence of the effectiveness and robustness of our GTSM module for all categories.

The MINS module effectively leverages complementary information provided by both modalities to mitigate the impact of intrinsic defects in each modality, resulting in notable performance improvements. It achieves an OA of 86.0%, Kappa of 80.3%, mIoU of 67.0%, and mF1 of 79.2%. Moreover, when the integrated feature extraction network incorporates GTSM in the optical branch and MINS in the SAR branch, a significant enhancement is observed, with an OA of 87.0% and Kappa of 81.7%. This surpasses the baseline model with further improvement of 5.5% in OA and 8.0% in Kappa. The fine-grained results in Fig. 6 and the visualizations in Fig. 7 effectively demonstrate the effectiveness of the MINS module in dealing with modality-intrinsic noise and false information.

Effectively fusing complementary information from different modalities is another crucial point for improving MSS performance. The AMFF module incorporates a novel idea to integrate useful information fully, achieving OA of 85.0%, Kappa of 78.8%, mIoU of 64.3%, and mF1 of 76.8%. The feature maps generated after incorporating the AMFF module (shown in Fig. 7) once again demonstrate the superiority of our proposed fusion strategy. It deals with feature mismatch, suppresses redundant information and modality-intrinsic noise, and fully exploits the complementary information explored in both modalities. Combining the AMFF module with either the GTSM or the MINS also leads to substantial improvements. Finally, our method outperforms the baseline model and achieves superior performance, with OA of 87.5% and Kappa of 82.5%, when all three components are employed in a cooperative manner. Notably, our approach achieves a significant improvement of approximately 14.9% in mIoU, providing compelling evidence of its efficacy and robustness for the semantic segmentation task. According to the visualizations in Fig. 7, our approach combined with the three modules achieves more satisfactory segmentation results with higher continuity and more precise edges.

4) Ablation Analysis for the Node in GTSM Module: To show the effectiveness of our choice for an important

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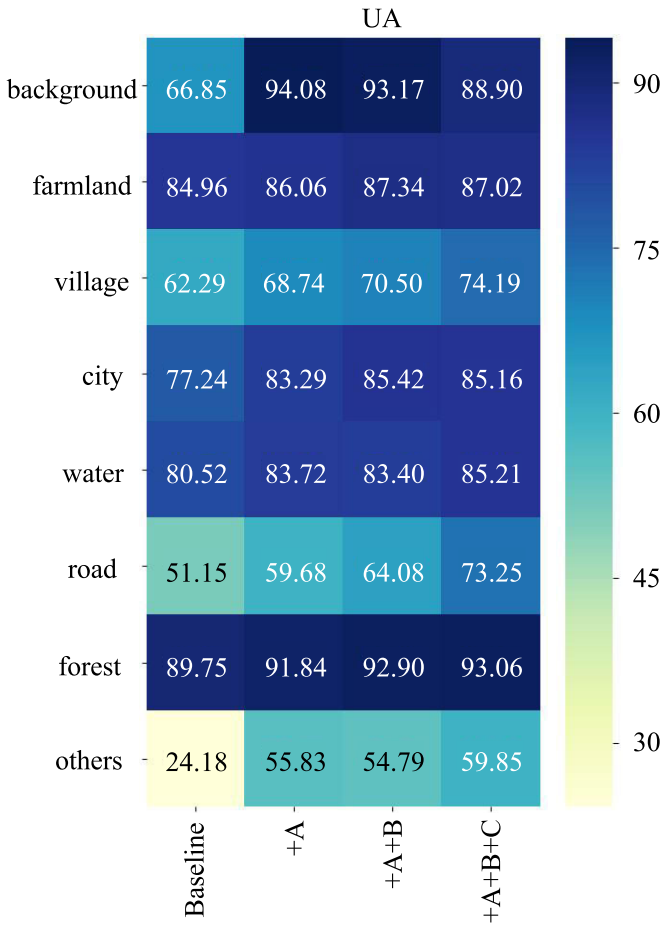


Fig. 6. Ablation studies of eight fine-grained categories with UA on the WHU-OPT-SAR validation set. "A," "B," and "C" correspond to the GTSM, MINS, and AMFF modules, respectively. We gradually add them based on the baseline model to evaluate the effectiveness of the three components. The darker the color, the higher the UA.

TABLE IV
ABLATION ANALYSIS FOR THE NUMBER OF GRAPH NODES

Method	Manner	nodes	OA(%)	Kappa(%)	mIoU(%)	mF1(%)
Baseline	-	-	81.5	73.7	54.2	68.1
Ours	fixed	64,64,64,64	86.8	81.5	68.4	80.3
		32,32,32,32	87.0	81.7	68.7	80.6
	decreasing	64,32,16,8	**87.5**	**82.5**	**69.1**	**81.1**
	increasing	8,16,32,64	86.8	81.5	68.0	80.1

hyperparameter, the number of nodes N, we perform some ablation experiments. Inspired by ResNet [67] and Swin transformer [32], [33], we design the GTSM structure with a hierarchical structure and compare three ways of the number of nodes: fixed, decreasing, and increasing. The detailed experimental results are presented in Table IV. As we all know, the shallow layers of the network pay more attention to some details, such as edges and textures. As the network deepens, features will encompass more and more semantic information. Therefore, the decreasing manner gradually reduces the number of nodes with the deepening of the network makes sense. Consequently, each node represents a larger receptive field, aligning better with the characteristics of the network and yielding improved results. Therefore, we adopt this changing manner of nodes in our experiments to further improve segmentation performance.

TABLE V
ABLATION ANALYSIS FOR THE LOSS FUNCTIONS

Method	Seg Loss	KL	Aux Loss	OA(%)	Kappa(%)	mIoU(%)	mF1(%)
ours	✓			86.9	81.7	65.6	78.5
	✓	✓		87.2	82.0	69.3	81.1
	✓	✓	✓	**87.5**	**82.5**	**69.1**	**81.1**

TABLE VI
SPACE COMPLEXITY ANALYSIS

Module	Params	FLOPs	OA	mIoU
Baseline	50.3M	77.9G	81.5	54.2
+4*GTSM	+1.9M	+8.1G	86.0	67.1
+4*MINS	+12.1M	+11.2G	86.0	67.0
+4*AMFF	+1.2M	+1.3G	85.0	64.3

5) Ablation Study for the Loss Hyperparameters: We use multiple loss functions to guide the training procedure of our proposed framework. In order to achieve the best performance, we conduct some experiments to choose the value of each hyperparameters. The results are shown in Fig. 8. It can be seen that the choice $\alpha = 9$ and $\beta = 0.1$ achieves the best result for the loss functions. It is worth mentioning that we conduct three experiments for each choice of every hyperparameter to eliminate the effects of training instability.

6) Ablation Analysis for the Loss Functions: We analyze the influence of various forms of supervision and present the corresponding results in Table V. According to the results, we confirm that suitable supervision can help the model learn feature representations effectively. The first row shows that using standard cross entropy (CE) as the supervision for final segmentation prediction yields an OA of 86.9%, Kappa of 81.7%, mIoU of 65.6%, and mF1 of 78.5%. KL loss explicitly constrains the consistency of two modalities' similarity affinity maps, thereby improving the quality of fused features and further improving the accuracy of the segmentation. Moreover, the auxiliary loss can help accelerate the model convergence and further refine the results. It is worth noting that we employ standard CE for supervising the optical branch, while for the SAR branch, we utilize the focal loss to further improve the performance of hard examples. In conclusion, the combination of all the losses results in the best performance, achieving an OA of 87.5%, a Kappa coefficient of 82.5%, a mean IoU (mIoU) of 69.1%, and an F1 score of 81.1%.

7) Space Complexity Analysis: We conduct the space complexity analysis of our MoCG on the WHU-OPT-SAR dataset. We calculate the increment of parameters and flops when adding each module. The results are shown in Table VI. According to the statistics on the Params, compared to baseline model, when adding GTSM modules and AMFF modules, the parameters only increase 3.8% and 2.4%, respectively. The OA yields 86% and 85%, with an increase of 4.5% and 3.5%. Compared to the other two modules, MINS modules cost

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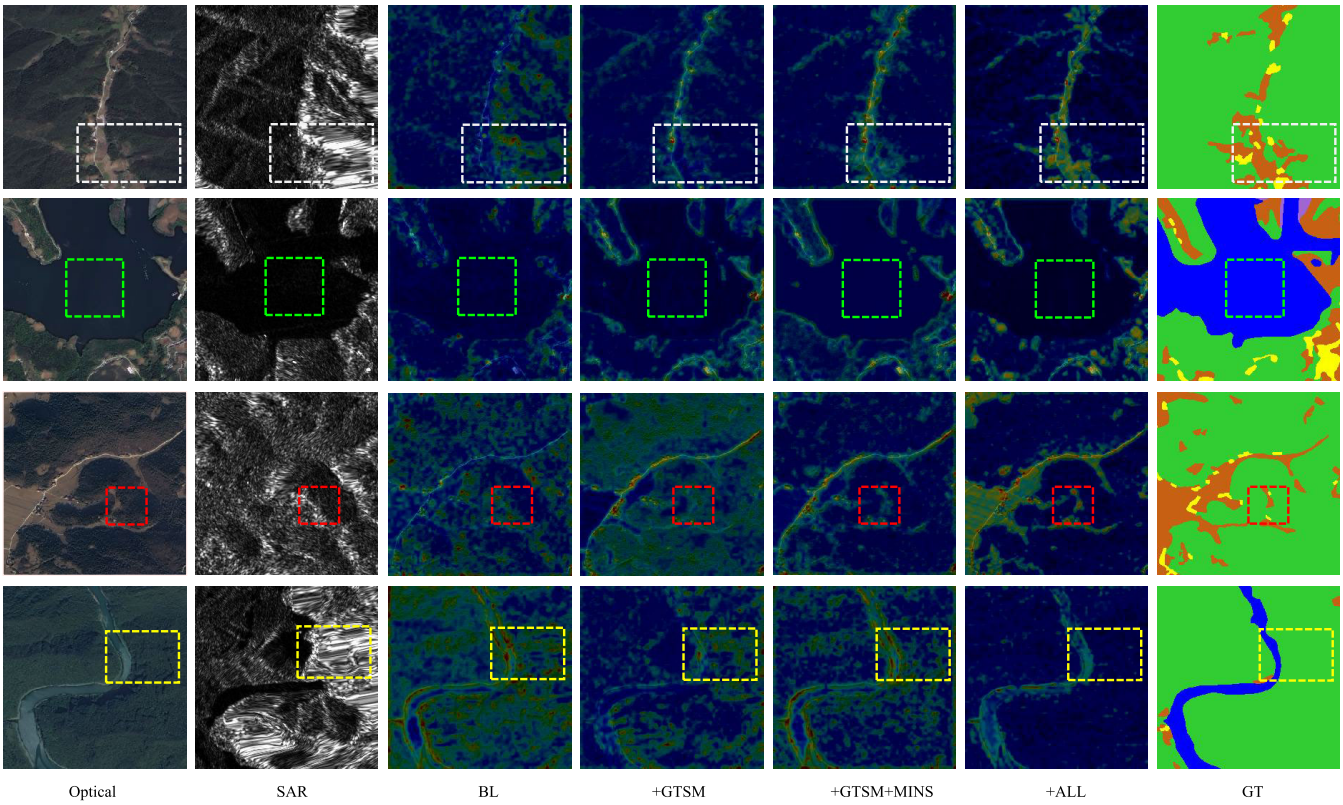


Fig. 7. Visualization feature maps of our ablation experiments compared to the baseline when adding different components, where "BL" stands for the baseline model, "GTSM" stands for hierarchical GTSM module, and "MINS" is on behalf of MINS module based on consistency of similarity. "ALL" means the model including all three components (GTSM, MINS, and AMFF), "GT" is the ground truth.

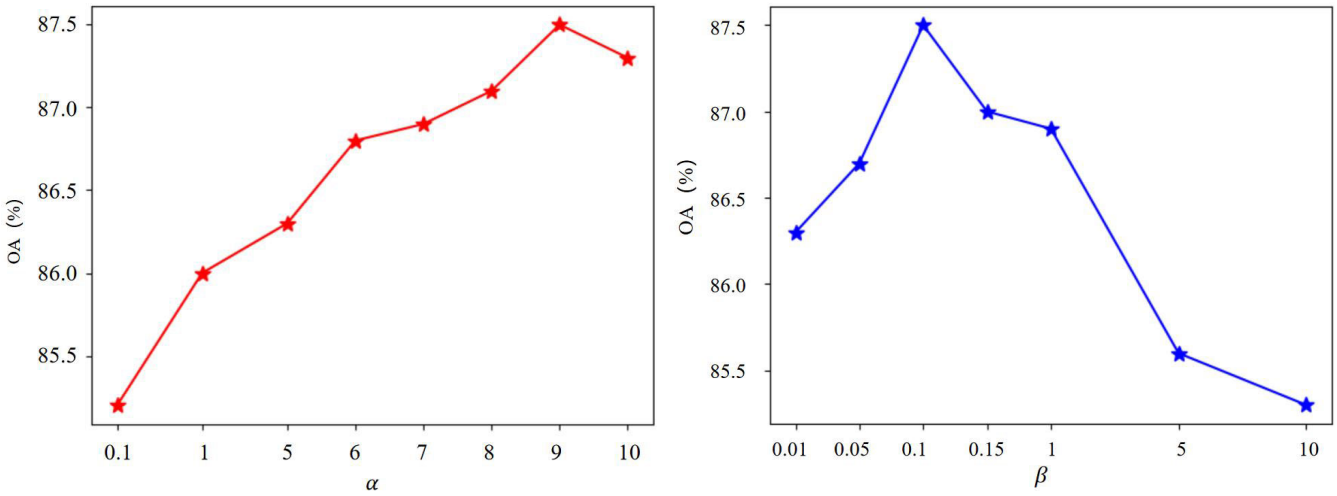


Fig. 8. Comparisons of different choices of loss hyperparameters α and β for auxiliary loss and KL loss.

much more and increase 24.1%. However, it also makes 4.5% improvements in OA. What's more, adding GTSM, MINS, and AMFF modules to the baseline model increases flops by 10.3%, 14.4%, and 1.7%, respectively.

8) Visualization Analysis: Fig. 9 displays some visualizations of baseline, other methods, and our proposed MoCG. Obviously, the baseline model exhibits fragmented and discontinuous segmentation performance. TokenFusion [35] demonstrates relatively good coherence in segmenting roads, but the performance remains unsatisfactory for small objects, narrow roads, and waters. TransFuser [36] tends to produce segmentation results with a circular tendency, lacking shape and distinct edges. MCANet [25] shows a significant amount of misclassifications, particularly for categories with high inter-class variations. In comparison, our approach has significantly improved the segmentation performance. Subjectively, there is a high inter-class continuity in all categories, especially in road and water. More specifically, the hierarchical GTSM structure helps our model fully explore the topological relationships between objects, maintaining the internal consistency

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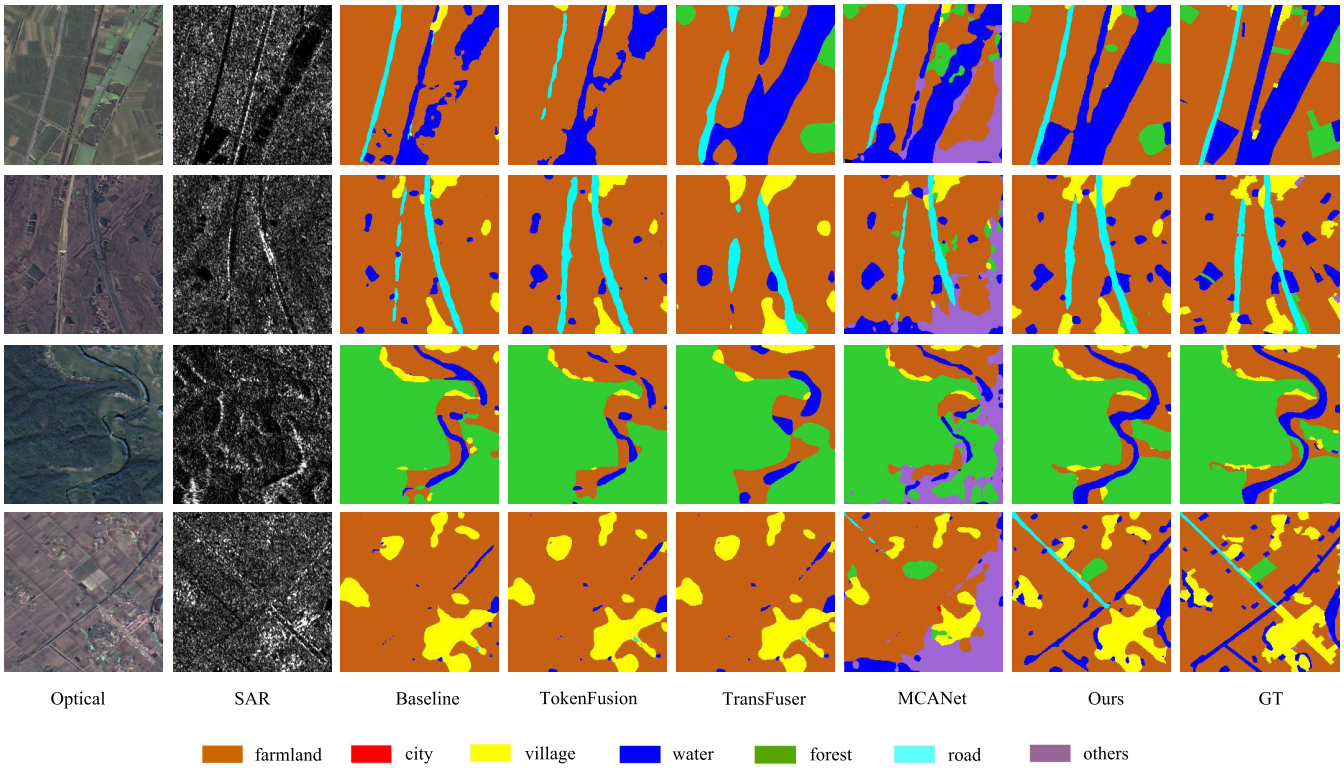


Fig. 9. Qualitative comparisons of different methods, baseline model and our MoCG framework on the WHU-OPT-SAR validation set.

of the foreground objects. Besides, our MINS module plays a crucial role in rectifying misleading information and mitigating the influence of inherent characteristics present in original images. Moreover, the AMFF module effectively harnesses the complementary features extracted from each modality while suppressing the impact of modality-intrinsic noise. This comprehensive fusion enables our proposed method to achieve superior results, highlighting its superiority over existing approaches.

E. Experiments on DFC23 Dataset

To assess the generalization capabilities of the proposed MoCG, we perform additional analysis on the DFC23 dataset.

1) Comparative Analysis With State-of-the-Art: In this section, we conduct a comprehensive comparison between our MoCG and some state-of-the-art approaches, encompassing both multimodal and single-modal approaches on the DFC23 dataset. Table VII reveals the performance. We have organized each section in ascending order according to the OA. From the first two sections of the table, it is evident that different methods exhibit varying performance depending on whether optical or SAR images are utilized. When using only optical images as input, CCNet [50] shows the best performance and yields OA of 93.8% and mIoU of 65.1%. While when using only SAR data as input, DANet [48] exhibits the best performance and achieves OA of 91.3% and mIoU of 60.6%. Different types of data have different properties, hence distinct methods demonstrate diverse performance. The third part of Table VII displays some multimodal results. MCANet [25] adopts a multimodal cross-attention module gains the best performance and achieves OA of 94.2% and mIoU of 67.9%. Although the segmentation results of optical images are significantly better than those of SAR, both are unsatisfactory. What's more, due to the lack of modality-specific feature extractors, these approaches cannot extract inherent characteristics for better image interpretation. Directly fusing features from two distinct modalities can even yield adverse effects, as different features exhibit significant disparities. Notably, the majority of multimodal segmentation methods do not achieve the same level of performance as single-modal segmentation methods that solely employ optical images as inputs, further underscoring this point. In contrast, our approach takes into account the modality-specific characteristics and designs separate feature extractors for optical and SAR images. Besides, we introduce a novel feature fusion module to mitigate the impact of the semantic gap between different modalities and suppress inherent defects in modes, yielding more favorable outcomes and further emphasizing the substantial contributions of our approach.

In a word, our proposed method surpasses other techniques and reaches a new state-of-the-art benchmark with remarkable achievements, achieving OA of 97.3% and mIoU of 86.7%. These outstanding results highlight the superior performance and effectiveness of our approach compared to existing methods.

2) Ablation Analysis for the Proposed Approaches: To certify the effects of each component on the DFC23 dataset, we carry out numerous ablation experiments about the GTSM, MINS, and AMFF modules in addition. According to the overall results shown in Table VIII, the baseline model achieves 91.8% OA, 76.8% Kappa, 47.4% mIoU, and 61.3% mF1. After adding the GTSM module, it makes a great improvement with

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TABLE VII
COMPARISON BETWEEN OUR METHOD AND OTHER SOTA METHODS ON DFC23 DATASET

Method	Modality	Backbone	IoU per category (%)													OA(%)	mIoU(%)
			BG	FR	GR	GR2	RR	MER	HRv1	HRv2	MR	PR	AR	**Do**	OR		
SETR [56]	OPT	ViT-L [52]	88.0	59.7	41.4	58.0	66.9	3.5	19.4	32.2	6.2	2.7	37.0	20.9	48.9	88.5	37.3
UperNet [75]	OPT	ResNet-50 [67]	92.2	60.2	46.6	44.9	62.9	11.3	8.6	32.2	7.4	25.8	44.3	59.9	30.2	90.3	40.5
DANet [48]	OPT	ResNet-50 [67]	90.9	72.1	58.0	59.1	82.7	45.3	50.1	53.7	65.9	30.3	74.8	76.2	70.3	91.9	63.8
Deeplabv3 [43]	OPT	ResNet-50 [67]	94.1	73.7	61.4	71.7	73.1	21.4	22.5	43.1	31.8	16.3	55.7	55.3	58.9	93.3	52.2
PSPNet [44]	OPT	ResNet-50 [67]	93.8	74.6	64.6	67.6	82.2	50.3	41.3	57.7	61.9	46.0	71.8	79.6	66.6	93.6	66.0
CCNet [50]	OPT	ResNet-50 [67]	94.0	75.2	65.5	70.2	71.3	46.5	37.0	57.1	55.3	43.9	74.1	78.2	67.6	93.8	65.1
UperNet [75]	SAR	ResNet-50 [67]	79.2	22.0	10.0	5.8	16.7	2.4	0.5	1.0	3.2	2.8	3.1	0.6	0.05	78.9	11.3
SETR [56]	SAR	ViT-L [52]	80.4	29.3	6.2	3.9	28.3	0.0	1e-5	1.1	0.07	0.0	7.3	0.0	8.1	80.9	12.7
Deeplabv3 [43]	SAR	ResNet-50 [67]	85.4	50.8	31.9	37.4	56.9	0.5	7.6	12.0	11.5	0.5	30.9	22.1	37.8	85.7	29.6
CCNet [50]	SAR	ResNet-50 [67]	86.2	54.7	37.9	45.6	69.1	12.8	14.6	23.3	28.3	0.9	49.1	51.1	51.0	87.0	40.3
PSPNet [44]	SAR	ResNet-50 [67]	86.5	56.1	40.9	50.1	72.1	24.1	30.6	31.8	46.6	11.7	60.4	64.2	56.7	87.6	48.6
DANet [48]	SAR	ResNet-50 [67]	90.2	71.2	56.8	57.5	81.4	38.4	47.9	51.8	63.1	13.9	72.6	73.7	69.4	91.3	60.6
FuseNet [20]	OPT&SAR	ResNet-50 [67]	82.4	36.0	19.1	18.6	42.8	3.4	2.6	7.8	2.3	5.6	16.2	6.6	11.9	82.0	19.6
CFNet [28]	OPT&SAR	MobileNetV2 [70]	84.2	44.7	22.8	31.2	41.1	0.03	5.6	6.8	0.0	5e-6	7.8	2.8	19.8	83.9	20.5
TransFuser [36]	OPT&SAR	ResNet-50 [67]	89.6	57.0	36.8	14.8	50.0	13.3	24.2	15.6	10.2	20.4	7.7	15.1	29.0	87.9	29.5
TokenFusion [35]	OPT&SAR	MiT-b1 [53]	90.2	55.5	40.9	43.2	56.9	12.4	24.6	17.0	9.8	22.6	13.2	15.3	18.2	88.9	32.3
PSCNN [18]	OPT&SAR	—	90.2	62.4	45.8	54.9	58.9	10.8	12.8	22.5	11.3	20.8	8.6	16.8	26.5	89.5	34.1
V_FuseNet [21]	OPT&SAR	ResNet-50 [67]	92.0	64.2	49.5	55.0	65.1	12.6	15.8	36.8	12.0	5.0	46.0	37.4	44.8	90.9	41.2
mmFormer [34]	OPT&SAR	—	91.7	67.7	58.8	72.2	76.3	20.3	42.0	62.1	16.8	21.5	57.0	5.2	50.3	91.4	49.4
MCANet [25]	OPT&SAR	ResNet-101 [67]	94.3	76.6	68.8	70.4	71.4	45.1	46.5	61.5	69.6	57.8	73.1	78.9	67.8	94.2	67.9
Ours	OPT&SAR	ResNet-50 [67]	96.4	87.1	83.4	85.0	92.2	72.8	78.3	81.5	80.0	71.2	86.0	84.6	87.0	96.8	83.5
Ours	**OPT&SAR**	**RegNet-3.2** [68]	**97.0**	**89.0**	**85.6**	**86.6**	**93.2**	**79.7**	**81.6**	**83.1**	**86.6**	**77.2**	**89.4**	**88.6**	**88.8**	**97.3**	**86.7**

* Note that the 12 foreground categories of DFC23 data set are: flat_roof (FR), gable_roof (GR), gambrel_roof (GR2), row_roof (RR), multiple_eave_roof (MER), hipped_roof_v1 (HRv1), hipped_roof_v2 (HRv2), mansard_roof (MR), pyramid_roof (PR), arched_roof (AR), dome (Do), others (OR).

TABLE VIII
ABLATION RESULTS OF AMFF, GTSM, AND MINS ON DFC23 DATASET

Method	Module			OA(%)	Kappa(%)	mIoU(%)	mF1(%)
	GTSM	MINS	AMFF				
Baseline				91.8	76.8	47.4	61.3
Ours	✓			94.1	83.5	66.6	79.3
		✓		93.7	82.9	64.7	77.4
			✓	94.2	84.3	62.8	75.4
	✓	✓		95.3	87.0	71.6	83.0
	✓		✓	95.2	86.9	70.6	82.3
		✓	✓	95.2	86.8	70.3	82.1
	✓	✓	✓	**97.3**	**92.9**	**86.7**	**92.8**

OA of 94.1%, Kappa of 83.5%, mIoU of 66.6%, and mF1 of 79.3%, surpassing the baseline by 2.3% in OA, 6.7% in Kappa, 19.2% in mIoU, and 18% in mF1. When integrating the MINS module, our model achieves OA of 93.7%, Kappa of 82.9%, mIoU of 64.7%, and mF1 of 77.4% and gains 1.9%, 6.1%, 17.3%, and 16.1% advancements in OA, Kappa, mIoU, and mF1. These two modules help the network to extract more modality-specific characteristics and deal with the intrinsic defects, resulting in significant improvement. Furthermore, when we integrate GTSM modules in the optical branch and MINS modules in the SAR branch simultaneously, the overall segmentation performance has been further improved. Specifically, OA has grown by 3.5%, Kappa has improved by 10.2%, and mIoU and mF1 gains 24.2% and 21.7% advancements compared to the baseline model, which further demonstrates the effectiveness of our modules. The AMFF module helps the model mitigate the influence of modality differences and suppress redundant and false information. Combining this module further makes greater performance with OA of 94.2%, Kappa of 84.3%, mIoU of 62.8%, and mF1 of 75.4%. What's more, another combination of the AMFF module with either the GTSM or the MINS module also makes sense, which demonstrates the three modules we proposed can mutually promote each other. Finally, our MoCG framework equipped with all components achieves new state-of-the-art results with OA of 97.3%, Kappa of 92.9%, mIoU of 86.7%, and mF1 of 92.8%, making a significant improvement over the baseline model.

In addition, we also make a comprehensive comparison on IoU of 12 fine-grained foreground categories and background when gradually adding three components. According to Fig. 10, our method significantly improves all categories compared with the baseline model, especially in PR, MR, MER, HRv1, and HRv2 categories. This proves our method is beneficial and robust for this task across all categories.

3) Visualization Analysis: We provide qualitative comparison on the DFC23 validation set, and the visualization results are shown in Fig. 11. The baseline model yields poor segmentation. TokenFusion [35] exhibits poor internal coherence and has a lot of misclassification pixels. TransFuser [36] produces more rounded edges without distinct corners, and it lacks the ability to effectively segment small objects. MCANet [25] achieves relatively better segmentation results, but it suffers from fragmented boundaries and also exhibits lower inter-class continuity. Compared with these methods, our approach achieves a more satisfactory

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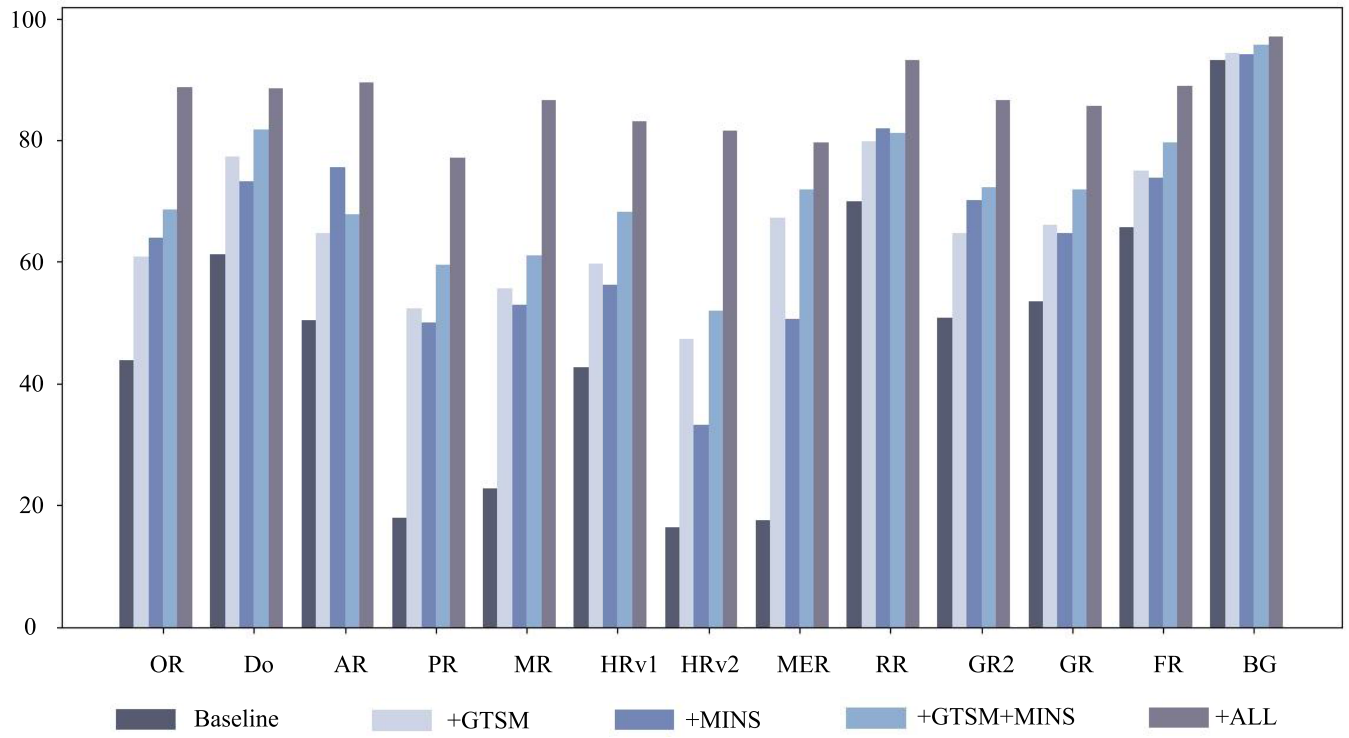


Fig. 10. Ablation studied of eight fine-grained categories with UA on the DFC23 validation set.

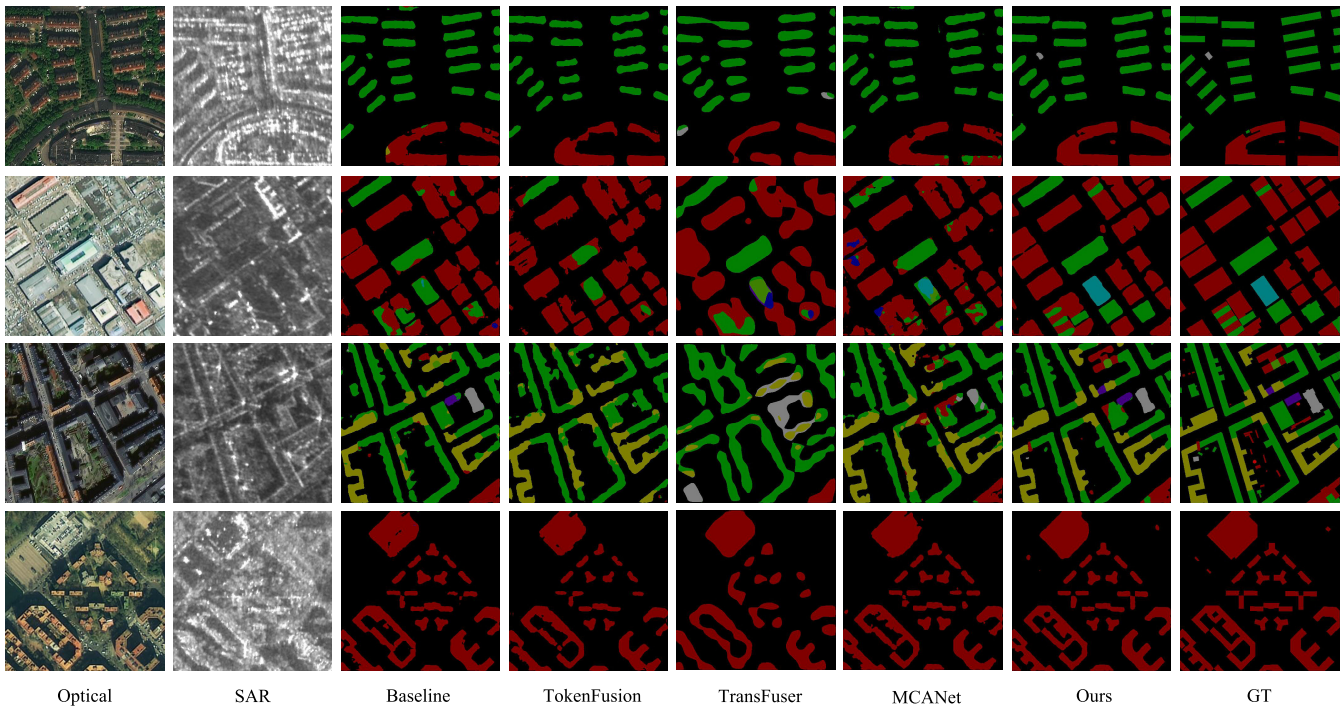


Fig. 11. Visualization comparisons of different methods, baseline model and our approach on the DFC23 validation set.

segmentation performance. The distinct backbones with GTSM and MINS modules explore the modality-specific characteristics and utilize the complementary information to compensate for the deficiencies, shortcomings, or limitations within each modality. More specifically, the GTSM module achieves higher inter-class consistency by exploring the geometric topological relationships between geographic objects. MINS module alleviates the false information caused by intrinsic modal properties. The AMFF module effectively integrates the valuable information provided by each

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modality by mitigating the impact of the semantic gap between modalities.

Overall, our method achieves higher accuracy, more complete object segmentation, clearer edge contours, and higher consistency within segmented objects, resulting in superior segmentation performance.

V. Conclusion

In this study, we devote to explore modality-specific characteristics and reduce the impact of semantic gaps between different modalities, effectively fusing the complementary information from both optical and SAR images. To this end, we propose the MoCG semantic segmentation in multimodal remote sensing images. This network incorporates the GTSM and MINS modules to serve as modality-specific backbones for heterogeneous data. What's more, we introduce the AMFF module into this framework, which leverages similarity consistency to maximize the utilization of complementary information for enhancing semantic segmentation accuracy. The evaluations on two datasets (WHU-OPT-SAR and DFC23) demonstrate the effectiveness, robustness, and generalization of our proposed approach.

Currently, we mainly focus on verifying the effectiveness of combining optical images and SAR images to improve the semantic segmentation performance, and verifying the feasibility of using graph networks to extract unique structural features from optical images. In the future, we will choose and design better graph modules to obtain higher segmentation accuracy. In addition, improving the efficiency of the model is also one of the directions of our research. What's more, we will take into account exploring the multitask multimodal tasks in remote sensing, particularly focusing on dense prediction tasks. As well, introducing more modalities is also one of our considerations. What's more, designing a universal model with prompt and more advanced technologies is also the direction of our exploration.

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